

Mathematics for Data Science - 1
End Sem Revision Session-2

1. If the piecewise function

$$f(x) = \begin{cases} mx & x < 3 \\ n & x = 3 \\ \underline{\underline{-2x + 9}} & x > 3 \end{cases} \quad \begin{matrix} (-\infty, 3) \\ \\ (3, \infty) \end{matrix}$$

is continuous, then the value of $m + n$ is

$$\begin{array}{ccc} \lim_{x \rightarrow 3^-} f(x) & = & \lim_{x \rightarrow 3^+} f(x) = f(3) = n \\ \parallel & & \parallel \\ 3m & & 3 \end{array}$$

$$\Rightarrow 3m = 3 = n.$$

$$\Rightarrow n = 3 \text{ and } m = 1$$

$$\Rightarrow m + n = 4.$$

2. Choose the correct options from the following.

$$f(x) = \underline{x^2 - \sin x}$$

2 > 0
 $|x^2| = x^2$
☒ The function $f(x) = |x^2 - 9| + \underline{|x|}$ is differentiable at $x = 3$. $\leftarrow$ at $\underline{x=3}$

☒ The function $f(x) = |x^2| - \sin(x)$ is differentiable at $x = 0$. $\rightarrow$

☐ The equation to the line tangent to the curve of $f(x) = x^3 + 2x + 3$ at the point $x = 1$ is $y - 5x - 1 = 0$.

☐ The equation to the line tangent to the curve of $f(x) = x^4 + x^2 + 5$ at the point $x = 1$ is $y - 6x - 1 = 0$.

$\leftarrow$ not diff at $\underline{x=3}$
• $f(x) = |x^2 - 9|$. $\leftarrow$ at $\underline{x=3}$

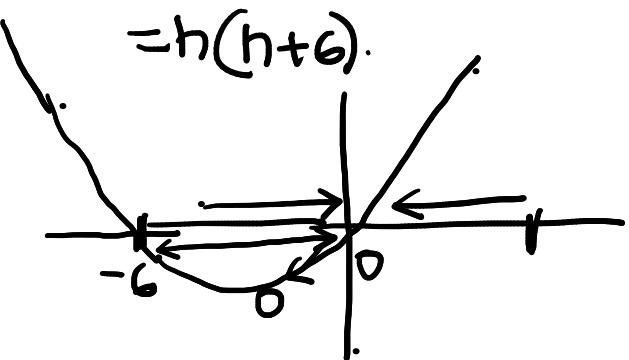
$$\lim_{h \rightarrow 0} \frac{f(3+h) - f(3)}{h}$$

$$|x| = -x$$

$$= \lim_{h \rightarrow 0} \frac{|(3+h)^2 - 9|}{h} = \lim_{h \rightarrow 0} \frac{|h^2 + 6h|}{h}$$

$$h^2 + 6h$$

$$= h(h+6)$$

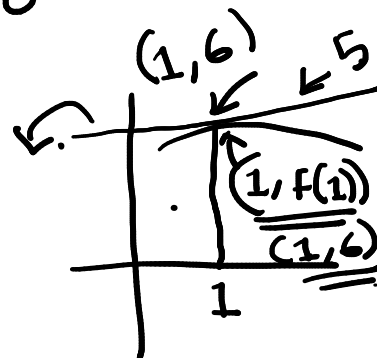


$$\lim_{h \rightarrow 0^-} \frac{-h^2 - 6h}{h} = -6$$

$$\lim_{h \rightarrow 0^+} \frac{h^2 + 6h}{h} = 6$$

• $\checkmark \leftarrow f(1) = \underline{6}$
 $f(x) = x^3 + 2x + 3$ at $x = 1$

$$\checkmark \underline{f'(x) = 3x^2 + 2} \Rightarrow \underline{f'(1) = 5}$$



3. Find local minima/maxima of the function $f(x) = x^3 - 3x + 2$.

1-Local minima

$$f''(x) = 6x.$$



$$f'(x) = 0 \Rightarrow 3x^2 - 3 = 0 \Rightarrow x = \pm 1$$

$$f'(1) = 6 > 0$$

$$f'(-1) = -6 < 0$$

Second der. : $f''(a) > 0 \Rightarrow a$ -Local min.

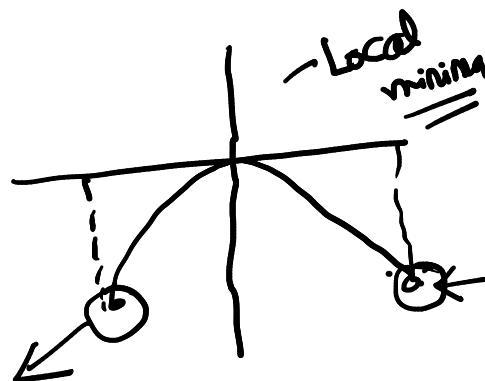
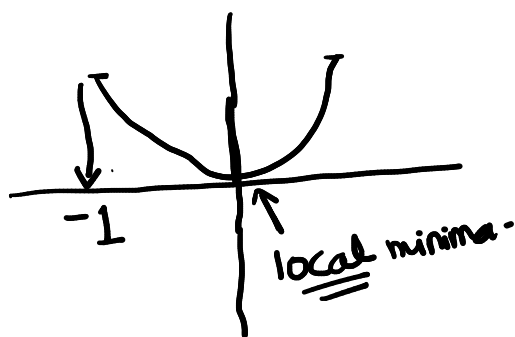
$f''(a) < 0 \Rightarrow$ " max.

$= 0 \Rightarrow$ Inclusive.

$$f(x) = x^2 \quad (-1, 1)$$

$\hookrightarrow$ Global maxima

$$\underline{\underline{f(x) = x^2 \text{ on } [-1, 1]}}$$



$$\underline{f: [-2, 3]}$$

$$3 + 4 - 12 + 1$$

end points

4. Find the global maxima/minima of $f(x) = 3x^4 - 4x^3 - 12x^2 + 1$ on the interval $[-2, 3]$.

$$f'(x) = 0$$

$$\Rightarrow x^3 - x^2 - 2x = 0$$

$$\Rightarrow x(x^2 - x - 2) = 0$$

$$\Rightarrow x(x+1)(x-2) = 0 \rightarrow \underline{\underline{0, -1, 2}}$$

$$\bullet f''(x) = 36x^2 - 24x - 24$$

$$f''(0) = -24 < 0 \rightarrow \text{local maxima}$$

$$\underline{f''(-1)} = 36 > 0 \rightarrow \text{local minima}$$

$$f''(2) > 0 \rightarrow \text{local } \underline{\underline{min}}$$

✓ Global maxima

$$f(0) = 1, f(-2) = \underline{\underline{33}}$$

$$f(3) = 28$$

✓ Global minima

$$f(-1) = -4, \underline{\underline{f(2) = -31}}$$

$$f(-2) = 33, f(3) = 28$$

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{f'(x)}{g'(x)}$$

5. Find $\lim_{x \rightarrow 0} x e^{\frac{1}{x}}$.

$\lim_{x \rightarrow \infty} \frac{\sin x + 1}{1}$

$\lim_{x \rightarrow 0} 1/x$

6. Find $\lim_{x \rightarrow \infty} \frac{x + \cos x}{x} = \lim_{x \rightarrow \infty} 1 + \frac{\cos x}{x} = 1$

$\lim_{x \rightarrow 0^+} x \cdot e^{1/x} = \lim_{x \rightarrow 0^+} \frac{e^{1/x}}{1/x} = \frac{\infty}{\infty}$

$= \lim_{x \rightarrow 0^+} \frac{e^{1/x} \cdot -1/x^2}{-1/x^2} = \lim_{x \rightarrow 0^+} e^{1/x} = +\infty$ (R.H.L)

$\lim_{x \rightarrow 0^-} x \cdot e^{1/x} = 0$ (L.H.L)

$\frac{1}{x} \rightarrow -\infty$

$e^{1/x} \rightarrow 0$

$0 \leq \frac{-1}{x} \leq \frac{\cos x}{x} \leq \frac{1}{x} \rightarrow 0$

$\lim_{x \rightarrow \infty} \frac{x + \cos x}{x} = \lim_{x \rightarrow \infty} 1 + \frac{\cos x}{x} = 1$

$0 \leq \frac{-1}{x} \leq \frac{\cos x}{x} \leq \frac{1}{x} \rightarrow 0$

Let (a, b) denote the point on the line $y = 2x + 3$ which is at a min. distance from the origin.

$b = y = 2x + 3$

$(x, y) \longleftrightarrow (0, 0)$ Parabola

$x^2 + y^2 = x^2 + (2x + 3)^2$ derivative
 $= x^2 + 4x^2 + 9 + 12x$
 $= 5x^2 + 12x + 9$

Value $x = -b/2a$

$$a = x = -\frac{b}{2a} = -\frac{-12}{10} = \frac{6}{5} \quad y =$$

- ✓ 7. Let $f(x) = 3x + 1$ then find the value of the integral $\int_0^2 f(x) dx$ using limit of Riemann sums as $n \rightarrow \infty$, for the given partition $P = \{0 = x_0, x_1 = \frac{2}{n}, \dots, x_i = \frac{2 \times i}{n}, \dots, x_n = 2\}$, $i = 1, 2, \dots, n$ and $x_i^* \in [x_{i-1}, x_i]$, where $x_i^* = \frac{2 \times i}{n}$.

→ $\int_0^2 f(x)$

$$\left[0, \frac{2}{n}\right], \left[\frac{2}{n}, \frac{4}{n}\right], \dots, \left[\frac{2 \times i}{n}, \frac{2 \times (i+1)}{n}\right], \dots, \left[\frac{2(n-1)}{n}, 2\right]$$

$\downarrow \quad \quad \quad \nwarrow \quad \quad \quad \nwarrow$
 $x_1^* \quad \quad \quad x_2^* \quad \quad \quad x_n^*$

$$f(x) = 3x + 1$$

$$\sum_{i=1}^n f(x_i^*) (\Delta x_i) = \sum_{i=1}^n f\left(\frac{2 \times i}{n}\right) \times \frac{2}{n}$$

$$= \sum_{i=1}^n \left(\frac{3 \times 2 \times i}{n} + 1 \right) \frac{2}{n}$$

$$= \sum_{i=1}^n \frac{12}{n^2} \times i + \sum_{i=1}^n \frac{2}{n}$$

$$= \frac{12}{n^2} \left(\sum_{i=1}^n i \right) + \frac{2}{n} \left(\sum_{i=1}^n 1 \right) \quad \leftarrow \quad \underbrace{1+1+\dots+1}_{n \text{ times}}$$

$$1+2+\dots+n = \frac{n(n+1)}{2}$$

$$= \frac{12}{n^2} \times \frac{n(n+1)}{2} + \frac{2}{n} \times n$$

as $n \rightarrow \infty$

$$= 6 + 2 = 8$$

$$\frac{12n^2 + 12n}{2n^2} = \frac{12 + 12/n}{2} \xrightarrow{n \rightarrow \infty} \frac{12}{2} = 6$$

$$\frac{12}{2} = 6$$

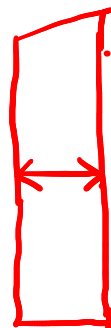
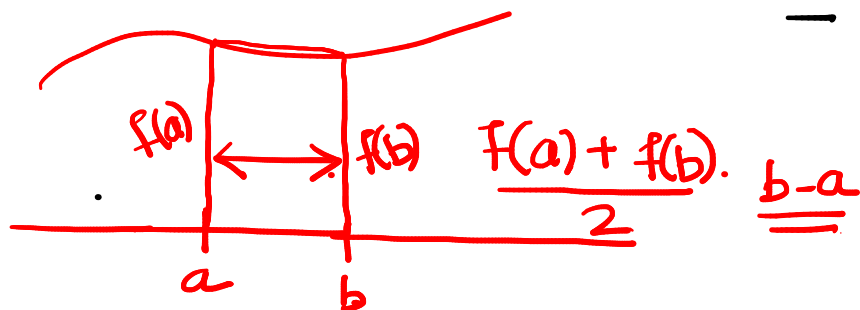
$$[x_{i-1}, x_i]$$

8. If the estimated area A of the region bounded by the graph of the function $f(x) = x^2 + 1$ above the interval $[1, 4]$ using the Left Riemann sum, by taking 3 subintervals of equal length. Find the value of A .

$$\frac{4-1}{3} = 1$$

$$\begin{matrix} a_1 & b_1 & a_2 & b_2 & a_3 & b_3 \\ \textcircled{1} & \textcircled{2} & \textcircled{2} & \textcircled{3} & \textcircled{3} & \textcircled{4} \end{matrix}$$

$$\frac{f(1)+f(2)}{2} \times 1 + \frac{f(2)+f(3)}{2} \times 1 + \frac{f(3)+f(4)}{2} \times 1$$



$$(0.913, \infty)$$

$$\log(x^3 + 5x^2 - x - 4)$$

9. Evaluate $\int e^x \sin(x) dx$ $\leftarrow$ by parts.

ILATE

$\swarrow$

$-\sin^{-1}x$

I: Inverse trig funct.

L: $\ln(x)$

A: $x, x^2, 1,$

T: $\sin x, \cos x$

E: e^x, a^x

$$\textcircled{=} \int u v dx = u \int v dx - \int u' (\int v dx) dx$$

$$\int \overset{v}{e^x} \overset{u}{\sin x} dx = \sin x \int e^x dx - \int \cos x e^x dx$$

$$= \sin x \cdot e^x - \cos x \cdot e^x - \int e^x \sin x dx$$

$$\int \overset{u}{\cos x} \overset{v}{e^x} dx = \cos x e^x + \int \sin x e^x dx$$

$$2 \int e^x \sin x dx = \sin x \cdot e^x - \cos x e^x$$

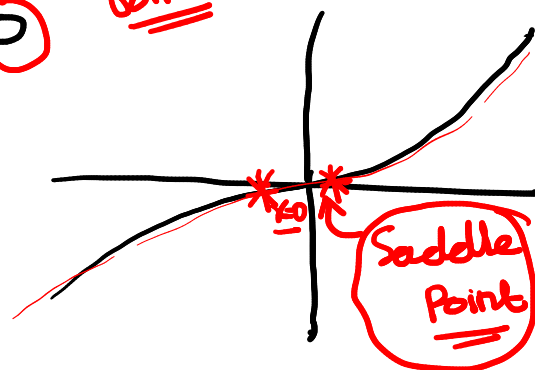
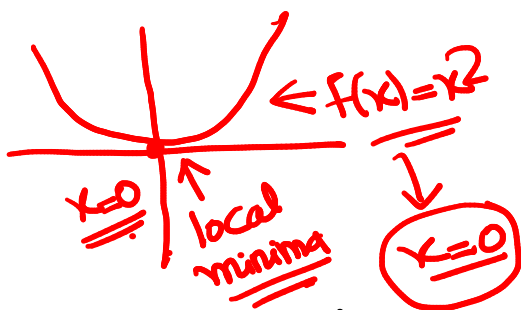
$$\Rightarrow \int e^x \sin x dx = \frac{e^x}{2} (\sin x - \cos x)$$

$$f(x) = x^3 \rightarrow$$

$$f'(x) = 0 \Rightarrow$$

$$\underline{\underline{x=0}}$$

$\rightarrow$ only critical point



10. Evaluate $\int x^2 e^x dx$. $u = x^2$ $v = e^x$

$$\lim_{x \rightarrow 0} x e^{1/x} = \lim_{x \rightarrow 0} \frac{e^{1/x}}{1/x}$$

$$\lim_{x \rightarrow 0^+} x e^{1/x} = \lim_{x \rightarrow 0^+} \frac{e^{1/x} \rightarrow +\infty}{1/x \rightarrow +\infty} = +\infty$$

$$\lim_{x \rightarrow 0^-} x e^{1/x} = 0$$

$$-x/2 = 2x + 3$$

$$\Rightarrow 2x - x/2 = 3$$

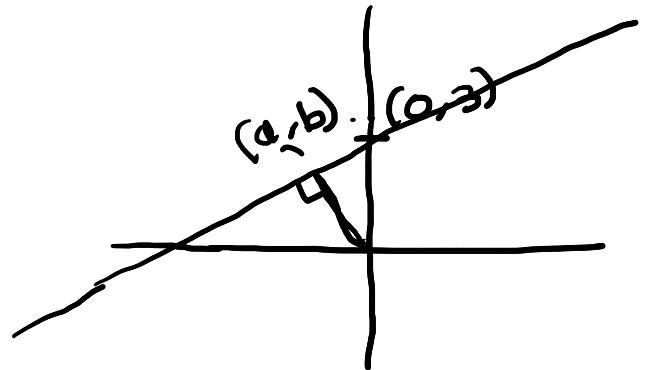
$$y = 2x + 3$$

$$\Rightarrow x = 6/5$$

$$m = 2$$

$$-1/2$$

$$y = -\frac{x}{2}$$



$$\int \frac{1}{\sqrt{1-x^2}} dx$$

$$\int_0^5 2x \sqrt{1+x^2} dx$$

$$\int_1^{26} \sqrt{t} dt$$

$$t = 1 + x^2$$

$$\frac{dt}{dx} = 2x \Rightarrow dt = 2x dx$$

$$x=0$$

$$t=1$$